

Exercise Sheet 3 - Solutions

Propositional Logic – Syntax & Natural Deduction

1.
 - “If I eat fries then I don’t eat an ice cream” is $f \rightarrow \neg i$
 - “If I eat a steak without fries then I eat it with peas” is $(s \wedge \neg f) \rightarrow p$
 - “If I eat chicken with peas then I also eat some fries” is $(c \wedge p) \rightarrow f$
 - “I never eat both steak and chicken at the same time” is $\neg(s \wedge c)$
 - “It is not true that I never eat steak and fries at the same time” is $\neg\neg(s \wedge f)$
2. One way to formalize the rules of the simple Sudoku puzzle is as follows:

$$\begin{aligned}
 &(p_0 \vee p_1) \wedge (q_0 \vee q_1) \wedge (r_0 \vee r_1) \wedge (s_0 \vee s_1) \\
 &\wedge (\neg p_0 \vee \neg p_1) \wedge (\neg q_0 \vee \neg q_1) \wedge (\neg r_0 \vee \neg r_1) \wedge (\neg s_0 \vee \neg s_1) \\
 &\wedge (p_0 \vee q_0) \wedge (r_0 \vee s_0) \\
 &\wedge (p_1 \vee q_1) \wedge (r_1 \vee s_1) \\
 &\wedge (p_0 \vee r_0) \wedge (q_0 \vee s_0) \\
 &\wedge (p_1 \vee r_1) \wedge (q_1 \vee s_1)
 \end{aligned}$$

where the 1st line expresses that each cell must contain either a 0 or a 1; the 2nd line expresses that each cell cannot contain both a 0 and a 1; the 3rd line expresses that each line must have a 0; the 4th line expresses that each line must have a 1 the 5th line expresses that each column must have a 0; the 6th line expresses that each column must have a 1.

3. Here is a proof of $A \rightarrow B \rightarrow A$:

$$\frac{\frac{\overline{A}^1}{B \rightarrow A}^2 [\rightarrow I]}{A \rightarrow B \rightarrow A}^1 [\rightarrow I]$$

4. Here is a proof of $(A \wedge B) \rightarrow \neg A \rightarrow C$

$$\frac{\frac{\frac{\overline{A \wedge B}^1}{A} [\wedge E_L] \quad \overline{\neg A}^2}{\perp} [\neg E]}{\frac{\overline{C}^1}{\neg A \rightarrow C}^2 [\rightarrow I]} [\perp E]$$

$$\frac{\overline{\neg A \rightarrow C}^2 [\rightarrow I]}{(A \wedge B) \rightarrow \neg A \rightarrow C}^1 [\rightarrow I]$$

5. Here is a proof of $(A \rightarrow B) \rightarrow (C \rightarrow \neg B) \rightarrow C \rightarrow \neg A$:

$$\frac{\frac{\frac{\overline{A \rightarrow B}^1}{B} [\rightarrow E] \quad \overline{A}^4}{\perp} [\neg E] \quad \frac{\overline{C \rightarrow \neg B}^2}{\neg B} [\rightarrow E]}{\frac{\overline{\neg A}^4}{\neg A} [\neg I]} [\neg E]$$

$$\frac{\frac{\overline{C \rightarrow \neg A}^3 [\rightarrow I]}{(C \rightarrow \neg B) \rightarrow C \rightarrow \neg A}^2 [\rightarrow I]}{(A \rightarrow B) \rightarrow (C \rightarrow \neg B) \rightarrow C \rightarrow \neg A}^1 [\rightarrow I]$$

6. The language is now

$$P ::= a \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \neg P \mid P \leftrightarrow P$$

To reason about if-and-only-if formulas, we can for example add the following introduction rule to Natural Deduction:

$$\frac{\begin{array}{c} \overline{P}^1 \\ \vdots \\ Q \end{array} \quad \begin{array}{c} \overline{Q}^1 \\ \vdots \\ P \end{array}}{P \leftrightarrow Q}^1 [\leftrightarrow I]$$

As an alternative to the above rule, we could instead have added the following introduction rule:

$$\frac{P \rightarrow Q \wedge Q \rightarrow P}{P \leftrightarrow Q} [\leftrightarrow I]$$

In addition we can add the following elimination rules:

$$\frac{P \leftrightarrow Q \quad P}{Q} [\leftrightarrow E_1] \quad \frac{P \leftrightarrow Q \quad Q}{P} [\leftrightarrow E_2]$$

As an alternative to the above elimination rules, we could instead have added the following rules:

$$\frac{P \leftrightarrow Q}{P \rightarrow Q} [\leftrightarrow E_1] \quad \frac{P \leftrightarrow Q}{Q \rightarrow P} [\leftrightarrow E_2]$$